QMD-Window Retraining Plan and Results B115T, Target 3deg

SMSimulator Reconstruction

April 20, 2026

Retraction / Update — 2026-04-20

Contamination found in original qmd-window training

- The original `formal_B115T3deg_qmdwindow/20260227_223007/` training consumed Geant4 output produced while the `DeutDetectorConstruction::fSetTarget` default was silently `true`. Tree-gun input (already post-IMQMD secondaries) was therefore scattered a second time by the Sn target.
- Audit: `docs/audit/2026-04-20-nn_target_contamination_audit.md`.
- The polluted formal run and all 7 `optimization_runs/*` sweeps were deleted; the Results / Sweep / Best-Candidate / Final-Accuracy tables in this deck therefore no longer reflect production state.

Clean retrain:

`data/nn_target_momentum/formal_B115T3deg/20260420_184345/`
(same narrow sampling: $R = 100$ MeV/c, $p_z \in [560, 680]$ MeV/c)

Val RMSE (MeV/c)	Contaminated	Clean retrain	Change
p_x	13.60	**6.70**	−51%

Motivation

- Task: reconstruct target proton momentum (p_x, p_y, p_z) from PDC two-point track.
- Existing formal model (wide sampling) is usable, but dataset efficiency is low.
- Baseline campaign: 20260227_205732.

Baseline key numbers

- Sampling: $p_\perp \leq 150$ MeV/c, $p_z \in [500, 700]$ MeV/c.
- Effective keep fraction (after PDC track reconstruction cuts): $\sim 24\%$.
- Momentum-norm RMSE: val 9.19 MeV/c, test 9.84 MeV/c.

Retraining Campaign Definition

Geometry and setup

- Geometry macro: `configs/simulation/DbeamTest/detailMag1to1.2T/geometry_B11`
- Target transform in run macro: $(-1.2488,\ 0.0009,\ -106.9458)$ cm, angle 3.0°.

New sampling window

- $p_\perp \leq 100$ MeV/c (disk in p_x, p_y).
- $p_z \in [560, 680]$ MeV/c.

Split size and seeds

- train/val/test = 200000/30000/30000.
- Seeds: 20260301 / 20260302 / 20260303.

Pipeline

1. Generate tree-input ROOT with proton truth sampled in new momentum window.
2. Run Geant4 simulation for train, val, test independently.
3. Build dataset CSV/ROOT with PDC track reconstruction.
4. Train MLP: $6 \to 128 \to 128 \to 64 \to 3$.
5. Evaluate val/test with independent hold-out files.

Tools

- `run_formal_training.sh`
- `generate_tree_input_disk_pz.C`
- `build_dataset.C`, `train_mlp.py`, `infer_mlp.py`, `evaluate_mlp.py`

MLP Architecture

Network: $6 \to 128 \to 128 \to 64 \to 3$

- Input (6): `pdc1_x, pdc1_y, pdc1_z, pdc2_x, pdc2_y, pdc2_z`
- Three hidden layers with **ReLU** activations: 128, 128, 64 units.
- Output (3): $(p_x, p_y, p_z)_{\text{truth}}$ in MeV/c (no output activation).

Evaluation metrics (post-training)

- Per-component RMSE, MAE, bias, and p_{95} for p_x, p_y, p_z.
- Momentum-norm RMSE: $\sqrt{\langle(|\hat{\vec{p}}| - |\vec{p}|)^2\rangle}$.

Loss Function and Optimiser

Training loss: MSE over all three components

$$\mathcal{L} = \frac{1}{3N} \sum_{i=1}^{N} \left[(\hat{p}_x^{(i)} - p_x^{(i)})^2 + (\hat{p}_y^{(i)} - p_y^{(i)})^2 + (\hat{p}_z^{(i)} - p_z^{(i)})^2 \right]$$

Implemented as `nn.MSELoss()` in PyTorch.

Optimiser and hyperparameters

- Optimiser: **Adam**, lr $= 10^{-3}$, weight decay 10^{-5}.
- Batch size: 1024; Max epochs: 120.
- **Early stopping**: patience = 12 epochs on validation loss; best-checkpoint model is restored at the end.

Data Preprocessing and Split

Feature normalisation (inputs only)

- z-score standardisation using training-set mean and std: $\tilde{x} = (x - \mu_{\text{train}})/\sigma_{\text{train}}$.
- Labels (p_x, p_y, p_z) are *not* normalised.
- Mean/std saved in `model_meta.json` for inference.

Dataset split (external files, fixed seeds)

- train / val / test = 200 000 / 30 000 / 30 000 events.
- Seeds: 20260301 / 20260302 / 20260303.
- Val loss used for model selection; test set held out until final evaluation.

Acceptance and Quality Criteria

Primary objective

- Improve effective kept samples by reducing out-of-acceptance phase space.

Evaluation criteria

- Keep fraction should improve from baseline $\sim 24\%$.
- Test $|\vec{p}|$ RMSE should remain stable or improve.
- Component-wise RMSE (p_x, p_y, p_z) tracked for regression.

Results [CONTAMINATED — see retraction frame]

Metric	Baseline (Wide)	QMD-Window (New)
Train kept fraction	23.61%	44.48%
Val kept fraction	23.14%	44.16%
Test kept fraction	23.44%	45.02%
Val $\lvert\vec{p}\rvert$ RMSE (MeV/c)	9.19	6.01
Test $\lvert\vec{p}\rvert$ RMSE (MeV/c)	9.84	5.88

RMSE numbers from the polluted model. Kept-fraction numbers (geometry acceptance only) are still valid. Clean retrain: val $\lvert\vec{p}\rvert$ RMSE $= 6.08$; see retraction frame for component-wise detail.

Post-Baseline Optimisation Objective

Why continue after qmd-window?

- QMD-window already improved usable sample rate and hold-out $|\vec{p}|$ RMSE.
- The next target is narrower: further reduce proton p_x/p_y reconstruction error on hold-out data.
- Project ranking metric for follow-up runs:

$$\mathrm{pxpy_rmse_mean} = \frac{\mathrm{RMSE}(p_x) + \mathrm{RMSE}(p_y)}{2}.$$

Search axes actually explored

- Manual loss weighting to emphasize p_x, p_y.
- Target z-score normalization.
- Wider MLP hidden dimensions.

Important comparison rule

- After target normalization, raw `best_val_loss` changes scale.
- Therefore cross-run ranking is based on raw hold-out RMSE metrics, not on loss value alone.

Optimisation Progress: Loss and Normalisation Sweep [CONTAMINATED]

All 7 sweep runs consumed the polluted dataset; the sweep was deleted and is not re-run — a clean re-sweep may pick a different winner.

Run	Val *pxpy* RMSE	Val p_z RMSE	Test $\lvert\vec{p}\rvert$ RMSE
Baseline qmd-window	13.0186	5.8863	5.8798
$2, 2, 1$	12.9918	5.9981	5.9780
$1.5, 1.5, 1$	12.9977	6.1136	6.1094
$3, 3, 1$	12.9235	5.9605	5.9649
$3, 3, 1$ + zscore	12.8254	5.4072	5.3718

Reading of the sweep

- Weighted loss alone helps, but gains remain modest.
- Target normalization produces the first clearly stronger jump in both *pxpy* and $\lvert\vec{p}\rvert$ metrics.

Current Best Candidate and Interpretation [CONTAMINATED]

Both rows of the table below are from the polluted dataset and are retained only to document the pre-audit state. **Reference vs current best training candidate**

Metric	Baseline qmd-window	Best current candidate
Configuration	128-128-64, MSE	256-256-128, 4, 3, 1+zscore
Val p_x RMSE	13.5973	13.3366
Val p_y RMSE	12.4398	12.2689
Val p_z RMSE	5.8863	5.3517
Val *pxpy* RMSE mean	13.0186	12.8027
Test $\lvert\vec{p}\rvert$ RMSE	5.8798	5.2488

Observed gain

- *pxpy* RMSE mean improves by about 1.66%.
- Test $\lvert\vec{p}\rvert$ RMSE improves by about 10.73%.

Data Flow (Target-Corrected Full-Event Campaign)

Execution chain

1. Reuse existing QMD-derived Geant4 input trees under: `data/simulation/g4input/y_pol/phi_random/...` and `data/simulation/g4input/z_pol/b_discrete/...`
2. Run campaign script: `scripts/analysis/run_sn124_nn_reco_pipeline.sh`
3. Script generates per-file Geant4 macros under: `data/simulation/g4output/eval_20260227_235751/_macros/*.mac`
4. Each generated macro executes: `configs/simulation/DbeamTest/detailMag1to1.2T/geometry_B11`
5. Geant4 output ROOT and NN reconstruction ROOT are written and then evaluated.

Where Artifacts Are Written

Geometry and model

- Geometry macro (target-fixed 3deg, `SetTarget false`):
  `configs/simulation/DbeamTest/detailMag1to1.2T/geometry_B11`
- **Clean production model (2026-04-20 retrain):**
  `data/nn_target_momentum/formal_B115T3deg/20260420_184345/mo`
  Previous path
  `formal_B115T3deg_qmdwindow/20260227_223007/model/` was deleted due to target contamination.

Campaign outputs

- Full Geant4 outputs:
  `data/simulation/g4output/eval_20260227_235751/`
- Full reconstruction outputs:
  `data/reconstruction/eval_20260227_235751/`
- Smoke-test outputs:
  `data/simulation/g4output/eval_target3deg_smoke_20260228_{yp`
  `data/reconstruction/eval_target3deg_smoke_20260228_{ypol,zp`

Final Accuracy (All Files, All Events) [INVALIDATED]

These `evaluate_reconstruct_sn_nn` RMSEs were produced by the now-deleted polluted model running over the clean `eval_20260227_235751/` reconstruction files. Matching efficiency (geometry acceptance) is unaffected; the RMSE rows must be re-derived with the 2026-04-20 clean model before quoting.

Metric	Overall	Y-pol	Z-pol
Files	156	16	140
Events total	5,413,747	5,178,328	235,419
Matched events	3,632,185	3,463,201	168,984
Reco efficiency (%)	67.0919	66.8787	71.7801
p_x RMSE (MeV/c)	38.7268	37.8827	53.1511
p_y RMSE (MeV/c)	11.2901	11.2611	11.8694
p_z RMSE (MeV/c)	11.9982	11.1051	23.8083
$\lvert\vec{p}\rvert$ RMSE (MeV/c)	8.4297	7.9028	15.7297

Reconstruction Effect

What changed after fixing target macro

- The campaign now runs with explicit target position and target angle in geometry macro.
- Full-event processing is enforced in production (`beamOn=0`, all selected files).
- Reconstruction yields high usable statistics: 3,632,185 matched proton events from 5,413,747 truth proton events (67.09% matching efficiency — geometry-only, unaffected by the model swap).

Current performance interpretation [pending re-eval]

- RMSE-based Y-pol / Z-pol comparisons depended on the polluted model and must be re-derived with the 2026-04-20 clean model.
- Validation-set comparison (from the clean retrain): p_x RMSE $13.60 \rightarrow 6.70$, p_y RMSE $12.44 \rightarrow 2.40$; full-event re-eval is expected to follow the same direction.
- Geometry chain and matching efficiency numbers remain production-grade.

Conclusion

- Narrowing to the qmd-window sampling significantly improves effective kept samples ($\sim 24\% \rightarrow \sim 45\%$) — this is a geometry-acceptance result and survives the contamination audit.
- The clean 2026-04-20 retrain on the same narrow window confirms the window choice: val $|\vec{p}|$ RMSE $= 6.08$ MeV/c with strong transverse gain (p_x -51%, p_y -81% vs the deleted polluted baseline).
- Clean production model: `formal_B115T3deg/20260420_184345/model/`.
- Follow-up: re-run `evaluate_reconstruct_sn_nn` over `eval_20260227_235751/` with the clean model; re-do the hyperparameter sweep on the clean dataset before claiming a best candidate.